Mayank Neupane

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Education

Bachelor of Engineering in Computer Engineering

Lassonde School of Engineering, York University | **GPA: 3.0/4.0**

Technical Skills

Programming: Python, C, C++, Java, SQL, Bash, PowerShell, Verilog, SystemVerilog

Hardware and Verification: Digital logic, embedded systems, PCB design, circuit testing, hardware troubleshooting, testbenches, Verilator, functional simulation

Systems and Networking: Windows, macOS, Linux (Ubuntu), PC assembly, system configuration, TCP/IP, DNS, DHCP, Active Directory basics

Tools: Git, GitHub Actions, VS Code, Microsoft 365, Zendesk, AnyDesk, TeamViewer, SQL databases, asset management

Technical Projects

Five-Stage RISC-V Processor Core

SystemVerilog, Python, Verilator, GitHub Actions | Source code

- Implemented a **32-bit RV32I processor** with a five-stage pipeline, **data forwarding**, **load-use hazard detection**, pipeline stalls, and branch flushing.
- Verified **17,848 retired instructions** against an **independent Python reference model**, checking instruction order, register results, and memory writes.
- Passed **48 benchmarks**, including 38 instruction tests and 10 full-program tests, plus **3 targeted regressions** for store forwarding, wrong-path ECALL handling, and pipeline completion on return.
- Automated simulator builds and benchmark and regression checks through **GitHub Actions**, producing verification reports and diagnostic logs.

Employee Performance Management System

C, SQL, Linux | Source code

- Developed a **modular C application** with employee CRUD operations, **dynamic memory allocation**, record search, and performance-based sorting.
- Benchmarked operations across **100–10,000 employee records**; measured median no-match search and sorting times of **0.458 ms** and **0.925 ms** for 10,000 records.
- Implemented **binary file persistence and SQL export**; measured median loading and export times of **0.130 ms** and **6.443 ms** for 10,000 records in local warm-cache benchmarks.

Work Experience

Laboratory Technician

Lassonde School of Engineering, York University | Oct. 2024 – May 2026

- Prepared engineering laboratories and provided technical support for **200+ students and faculty**, ensuring equipment and workstations were ready for scheduled sessions.
- Maintained inventory and lifecycle records for **150+ laboratory devices**, using **SQL databases** to track equipment allocation and asset status.
- Configured, maintained, and troubleshot **100+ computers**, embedded systems, and laboratory devices, resolving **hardware, software, and connectivity issues**.
- Installed **operating systems**, software environments, **device drivers**, and **network settings** to support engineering coursework and laboratory activities.
- Worked with faculty, teaching assistants, and technical teams to identify laboratory requirements and **resolve equipment failures** during instructional sessions.
- Documented technical incidents, maintenance activities, and **troubleshooting procedures** to support issue tracking and knowledge sharing.

Supervisor, Ride Department

Centreville Amusement Park | May 2023 – Sept. 2026

- Coordinated daily operations for **100+ ride operators** across multiple attractions, managing staff schedules, assignments, and operational coverage.
- Adjusted employee assignments in response to absences, scheduling conflicts, and **equipment disruptions** to maintain service coverage.
- Conducted routine **safety inspections**, identified operational risks, and coordinated corrective actions under established safety procedures.
- Prepared daily operational reports, **incident documentation**, safety inspection records, and employee performance reports for department management.
- **Trained and mentored operators** on ride procedures, safety protocols, and customer service, reinforcing consistent performance and team accountability.

Certifications Licences and Awards

- **Project Management Foundations** — LinkedIn Learning | In progress
- **Ontario Class G Driver's Licence**
- **Employee of the Season** — Centreville Amusement Park | **Awarded twice**